

DevOps Series

Creating a Virtual Machine for Erlang/OTP Using Ansible



This seventh article in the DevOps series is a tutorial on how to create a test virtual machine (VM) to compile, build, and test Erlang/OTP from its source code. You can then adapt the method to create different VMs for various Erlang releases.

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Erlang is a programming language designed by Ericsson primarily for soft real-time systems. The Open Telecom Platform (OTP) consists of libraries, applications and tools to be used with Erlang to implement services that require high availability. In this article, we will create a test virtual machine (VM) to compile, build, and test Erlang/OTP from its source code. This allows you to create VMs with different Erlang release versions for testing.

The Erlang programming language was developed by Joe Armstrong, Robert Virding and Mike Williams in 1986 and released as free and open source software in 1998. It was initially designed to work with telecom switches, but is widely used today in large scale, distributed systems. Erlang is a concurrent and functional programming language, and is released under the Apache License 2.0.

Setting it up

A CentOS 6.8 virtual machine (VM) running on KVM is used for the installation. Internet access should be available from the guest machine. The VM should have at least 2GB of RAM allotted to build the Erlang/OTP documentation. The Ansible version used on the host (Parabola GNU/Linux-libre x86_64) is 2.3.0.0. The *ansible/* folder contains the following files:

```
ansible/inventory/kvm/inventory
ansible/playbooks/configuration/erlang.yml
```

The IP address of the guest CentOS 6.8 VM is added to the *inventory* file as shown below:

```
erlang ansible_host=192.168.122.150 ansible_connection=ssh
ansible_user=bravo ansible_password=password
```

An entry for the *erlang* host is also added to the */etc/hosts* file as indicated below:

```
192.168.122.150 erlang
```

A 'bravo' user account is created on the test VM, and is added to the 'wheel' group. The */etc/sudoers* file also has the following line uncommented, so that the 'bravo' user will be able to execute sudo commands:

```
## Allows people in group wheel to run all commands
%wheel ALL=(ALL) ALL
```

We can obtain the Erlang/OTP sources from a stable tarball, or clone the Git repository. The steps involved in both these cases are discussed below.

Building from the source tarball

The Erlang/OTP stable releases are available at <http://www.erlang.org/downloads>. The build process is divided into many